

Bugfix Log

Revision: 15 **Last modified:** 2026-06-13T18:15:00Z

Per CONST-MD-Bugfix-Documentation, every bug surfaced during implementation gets a permanent entry below: title, root cause, affected files, fix description, regression guard.

Entries are append-only; do not edit historical entries except to add clarification footnotes.

2026-04-27 “boba-jackett implementation (plan 2026-04-27-jackett-management-ui-and-system-db.md)

1. Master-key two-step write silent data-loss window

Severity: HIGH (silent credential loss possible on crash mid-bootstrap).

Root cause: Initial implementation of `bootstrap.EnsureMasterKey` generated `BOBA_MASTER_KEY` in memory, returned it to the caller, and *then* wrote `.env` in a separate step. If the process crashed, was killed, or the host lost power between those two operations, the caller would proceed to encrypt credentials with a key that disk had no record of “making them permanently undecipherable.

Affected files: - `qBittorrent-go/internal/bootstrap/bootstrap.go` - `qBittorrent-go/internal/envfile/write.go`

Fix: Collapse the generate + write into a single `envfile.Atomic` write (`tmpfile + fsync + rename`) before the function returns. Caller never sees a key that hasn’t been durably persisted. Commit `c0f20bc`.

Regression guard: - `qBittorrent-go/internal/bootstrap/bootstrap_test.go::TestEnsureMasterKeyDoesNotDuplicateHeader` - `qBittorrent-go/internal/envfile/write_test.go` atomic-write test layer

2. SQLite database file mode 0644 (world-readable credentials)

Severity: CRITICAL (any local user could read encrypted credentials + ciphertext-only attack surface).

Root cause: `modernc.org/sqlite` creates database files with the process’s default `umask`, which on the container yielded `0644` (world-readable). The Go service’s `.env` was already `0600` via `envfile.Atomic`, but `boba.db` was leaking “a defense-in-depth breach since the rows are encrypted, but still a violation of the principle “the DB file should be readable only by the service owner”.

Affected files: - `qBittorrent-go/internal/db/conn.go`

Fix: After the first `Open()` succeeds, `db.Open` calls `os.Chmod(path, 0o600)` (and the `-wal / -shm` siblings if present), collapsing the file mode to owner-only. Commit `8405167`.

Regression guard: - Layer 4 security challenge challenges/scripts/boba_db_file_perms_challenge.sh - qBitTorrent-go/internal/db/credential_leak_test.go

3. AutoconfigResult nil slices marshalled as JSON null

Severity: MEDIUM (silent client-breaking schema drift; UI rendered "undefined" badges instead of "0 items").

Root cause: Go's json.Marshal serializes a nil slice as null, not []. When Autoconfigure() ran on a fresh DB with no discovered credentials, the discovered_credentials, configured_now, and already_present fields all marshalled as null. The OpenAPI 3.1 schema declared them as array, so clients (including the Angular dashboard) treated null as a contract violation.

Affected files: - qBitTorrent-go/internal/jackett/autoconfig.go

Fix: Pre-allocate empty slices ([]string{}) at the top of Autoconfigure() so a "nothing to do" run still serializes as { ..., "discovered": [], "configured_now": [], ... }. Commit 6f3dbaf.

Regression guard: - Layer 6 contract test qBitTorrent-go/tests/contract/openapi_test.go (validates each named field is array per OpenAPI schema even on empty runs).

4. Catalog ReplaceAll empty-input wipe risk

Severity: HIGH (would permanently wipe the indexer catalog if Jackett returned an empty response; UI would show no indexers and fuzzy matcher would have nothing to match against).

Root cause: repos.IndexerCatalog.ReplaceAll(rows []Row) unconditionally DELETED existing rows then INSERTed the new ones. If rows was empty (Jackett momentarily unhealthy, transient 502, or catalog parse failure), the delete still happened, leaving an empty catalog table. Next refresh would have nothing to compare against.

Affected files: - qBitTorrent-go/internal/db/catalog.go (or equivalent repo file)

Fix: Refuse the operation early with errors.New("repos: ReplaceAll refusing empty replacement"). The caller (autoconfig orchestrator) treats this as a soft error, logs it, and leaves the previous catalog intact. Commit 8d71df1.

Regression guard: - qBitTorrent-go/internal/jackettapi/catalog_test.go::TestRefreshCatalogAllTemplatesFailReturns200WithErrors

5. TestSearchHandler_QueueFull pre-existing flake (NOT a boba-jackett bug)

Severity: LOW (test-only; flagged for follow-up).

Status: NOT FIXED in this plan. Confirmed unrelated to boba-jackett work.

Location: qBitTorrent-go/internal/api/api_test.go

Symptom: Occasional intermittent failure under load when the search queue fills before the test's wait returns.

Action: Logged here for traceability so future audits don't mis-attribute it. Open issue suggestion: stabilise by injecting a deterministic queue-full hook instead of racing against real goroutine scheduling.

2026-06-09 "Session 7 fixes (env_loader, AsyncMock, gamestorrents)

6. env_loader flaky test: KEY2 leak across test ordering

Severity: LOW (test-only flake; not a production bug).

Root cause: test_comment_lines_ignored in tests/unit/test_env_loader.py asserted os.environ.get("KEY2") is None after calling load_env_files() with a config file where KEY2=commented_out was commented out. However, load_env_files has a "first wins" policy if KEY2 was already present in os.environ from a prior test's load_env_files call, the function would not override it. Under pytest-randomly ordering, the test that sets KEY2 sometimes ran before the comment-line test, causing intermittent failure.

Affected files: - tests/unit/test_env_loader.py

Fix: Added explicit os.environ.pop("KEY1", None) and os.environ.pop("KEY2", None) at the START of the test (before calling load_env_files), ensuring a clean env state regardless of test execution order. The finally block was retained as a safety net.

Regression guard: - tests/unit/test_env_loader.py::test_comment_lines_ignored ran 2147/2147 twice consecutively under random ordering with zero failures.

7. AsyncMock warning in search deep-coverage tests

Severity: LOW (test warning; not a production bug).

Root cause: test_iptorrents_login_no_cookies in tests/unit/merge_service/test_search_deep_coverage.py used AsyncMock() for login_resp and mock_session. These objects are used as context managers (async with session.post(...) as resp), not as directly awaited coroutines. AsyncMock.__aenter__ returns a coroutine, but the test's mock setup didn't properly wire __aenter__/__aexit__, causing "coroutine was never awaited" RuntimeWarning.

Affected files: - tests/unit/merge_service/test_search_deep_coverage.py

Fix: Changed AsyncMock() to MagicMock() with explicit __aenter__ and __aexit__ stubs, matching the actual usage pattern (context manager, not coroutine).

Regression guard: - tests/unit/merge_service/
test_search_deep_coverage.py â€” 0 AsyncMock warnings from this test (was 3).

8. gamestorrents _parse_size B-substring bug (documented, not fixed)

Severity: MEDIUM (plugin returns 0 for all realistic file sizes).

Root cause: Same class of bug as BOB-013 (torrentkitty). The `multipliers` dict in `_parse_size` iterates in insertion order: `{"B": 1, "KB": 1024, ...}`. Since "B" is a substring of "KB", "MB", "GB", and "TB", the check `if unit in size_str` matches "B" first for all realistic sizes. Then `size_str.replace("B", "")` leaves a trailing unit character (e.g. "35.2 G") which `float()` cannot parse, returning 0.

Affected files: - plugins/gamestorrents.py (line 74-86)

Fix: NOT FIXED in this session. Documented via tests that assert actual behavior (tests suffixed `_b_substring_bug`). Fix requires reordering the dict keys to check longest units first (same approach as BOB-013).

Regression guard: - tests/unit/
test_plugin_gamestorrents.py::TestParseSize â€” 8 tests documenting the bug across GB/MB/KB/TB/commas/uppercase.

2026-06-09 â€” Session 8 parallel plugin tests + gamestorrents fix

9. gamestorrents _parse_size B-substring bug (fixed)

Severity: MEDIUM (plugin returned 0 for all realistic file sizes).

Root cause: Same class of bug as BOB-013 (torrentkitty). The `multipliers` dict in `_parse_size` iterated in insertion order: `{"B": 1, "KB": 1024, ...}`. Since "B" is a substring of "KB", "MB", "GB", and "TB", the check `if unit in size_str` matched "B" first for all realistic sizes. Then `size_str.replace("B", "")` left a trailing unit character (e.g. "35.2 G") which `float()` could not parse, returning 0.

Affected files: - plugins/gamestorrents.py (line 74-86)

Fix: Reordered dict keys to check longest units first: `{"TB": ..., "GB": ..., "MB": ..., "KB": ..., "B": 1}`. Same approach as BOB-013.

Regression guard: - tests/unit/
test_plugin_gamestorrents.py::TestParseSize â€” 8 tests all pass with correct byte values for GB/MB/KB/TB/B/comma/uppercase.

10. piratebay import os placed after use (documented, not fixed)

Severity: LOW (only affects non-magnet torrent file downloads, which are rare).

Root cause: piratebay.py:175 has `import os` placed after `os.fdopen` on line 172. When downloading a non-magnet torrent file, `os.fdopen` is called before `os` is imported, causing `UnboundLocalError`.

Affected files: - plugins/piratebay.py (line 172-175)

Fix: NOT FIXED in this session. Documented via test `test_torrent_file_download_known_bug_os_import_order`. Fix requires moving `import os` to the top of the `download_torrent` method or module level.

Regression guard: - tests/unit/test_plugin_piratebay.py::TestDownloadTorrent::test_torrent_file_download_

11. torlock search() does not catch exceptions from retrieve_url()

Severity: LOW (network errors crash the entire search loop instead of failing gracefully).

Root cause: torlock.py search() calls retrieve_url() without a try/except, so a network error propagates up and crashes the search loop.

Affected files: - plugins/torlock.py

Fix: NOT FIXED in this session. Documented via test `test_search_exception_propagates`. Fix requires wrapping `retrieve_url()` in try/except with graceful error handling.

Regression guard: - tests/unit/test_plugin_torlock.py::TestSearch::test_search_exception_propagates

2026-06-09 â€” Session 8 wave 2: nyaa/kickass/anilibra/torrentgalaxy/yts

12. nyaa download_torrent missing import re (documented, not fixed)

Severity: MEDIUM (any nyaa.si URL passed to `download_torrent` raises `NameError`).

Root cause: plugins/nyaa.py:170 calls `re.search()` but `import re` is absent from the module. The `search()` method works because `re` is imported transitively via other modules, but `download_torrent` fails with `NameError: name 're' is not defined`.

Affected files: - plugins/nyaa.py (line 170)

Fix: NOT FIXED in this session. Documented via tests that assert the `NameError`. Fix requires adding `import re` at the top of the file.

Regression guard: - tests/unit/test_plugin_nyaa.py::TestDownloadTorrent â€” 6 tests including the `NameError` documentation.

13. kickass comma-separated size not matched by regex (documented, not fixed)

Severity: LOW (sizes like 1 , 234 . 5 MB silently parse to 0).

Root cause: The HTMLParser regex `\d+\ . \d+` doesn't match comma-separated numbers like 1 , 234 . 5 MB. The comma is not handled in the regex or the replacement logic.

Affected files: - `plugins/kickass.py`

Fix: NOT FIXED in this session. Documented via test `test_comma_in_size_not_matched_by_regex`. Fix requires updating the regex to handle commas.

Regression guard: - `tests/unit/test_plugin_kickass.py::TestHTMLParserFeed::test_comma_in_size_not_matched`

14. kickass crash-prone patterns (fixed)

Severity: MEDIUM (empty responses from `retrieve_url` crash the plugin).

Root cause: Three methods in `kickass.py` called `retrieve_url()` without try/except or empty-response guards: `__retrieve_download_link()`, `download_torrent()`, and `search()`. When `retrieve_url` returned None or empty string, subsequent `re.search/re.sub` calls crashed with `TypeError`.

Affected files: - `plugins/kickass.py` (lines 61, 74, 94)

Fix: Added try/except + empty-response guards to all three methods. Each now handles empty/None responses gracefully (returns `None`, prints fallback URL, or breaks the loop respectively).

Regression guard: - `tests/unit/test_plugin_kickass_guards.py` 13 tests proving empty response, None response, `ConnectionError`, `TimeoutError`, and malformed HTML don't crash.

2026-06-09 "Waves 5-8: B-substring epidemic across 9 community plugins

15. Systemic `_parse_size` B-substring bug (9 instances found, all fixed)

Severity: MEDIUM (all 9 plugins returned 0 for KB/MB/GB/TB "silently broken size reporting").

Root cause: The `_parse_size` method in each plugin uses a `multipliers` dict iterated in insertion order. With keys ordered `{"B": 1, "KB": 1024, ...}`, the check "B" in `size_str` matches the "B" substring inside "KB", "MB", "GB", and "TB". Then `size_str.replace("B", "")` leaves a trailing character (e.g. "1.5 G") which `float()` cannot parse, returning 0.

Affected plugins (9 total): | Plugin | Status | | gamestorrents.py | Fixed (BOB-024) | | megapeer.py | Fixed (BOB-042) | | one337x.py | Fixed (BOB-043) | | extratorrent.py | Fixed (BOB-044) | | torrentfunk.py | Fixed (BOB-045) | | therarbg.py | Fixed (BOB-047) | | pctorrent.py | Fixed (BOB-052, by subagent) | | bitru.py | Fixed (BOB-054) | | xfsusb.py | Fixed (BOB-057) | | yihua.py | Fixed (BOB-058) |

Fix: Reorder dict keys longest-unit-first: {"TB": ..., "GB": ..., "MB": ..., "KB": ..., "B": 1}. Same approach as BOB-013 (torrentkitty).

Global regression guard: 10 dedicated test files, each with explicit `_parse_size` assertions for all 5 units (B/KB/MB/GB/TB).

16. Systematic import re omissions (3 instances found, all fixed)

Root cause: Three plugins called `re.search()` or `re.compile()` without importing `re` at module level. The `search()` method worked via transitive imports from helper modules, but `download_torrent()` failed with `NameError`.

Affected plugins: | Plugin | Status | | nyaa.py | Fixed (BOB-035) | | audiobookbay.py | Fixed (BOB-042) | | (torlock `search()` also lacks exception handling â€” documented BOB-029) |

Fix: Added `import re` at module level.

17. bt4g infinite loop (test-only, fixed)

Severity: LOW (test hang, not production).

Root cause: `search()` has a `while True` loop that breaks when `parser.noTorrents` is `True`. Tests that used `return_value=MATCHING_HTML` (not `side_effect`) caused every page to match, triggering infinite loop.

Fix: Changed to `side_effect=[MATCHING_HTML, EMPTY_HTML]` so page 2 triggers the `noTorrents` break condition.

2026-06-13 â€” BobaLink extension: flaky-perf-test hardening + WCAG contrast fixes

Batch verified together by `extension/ci-ext.sh` â†’ CI-EXT: PASS, full suite **632 passed (632)**, `tsc --noEmit clean`, `npm run lint` 0/0, chrome+firefox builds loadable, Â§11.4.38 asset-verify pass, both store zips â‰¥10 KiB. Every fix below carries a permanent regression guard (Â§11.4.135) that was RED before the fix.

18. Junk-flood DoS security test â€” load-coupled absolute-wall-clock FAIL-bluff

Severity: MEDIUM (flaky test â€” Â§11.4.1 FAIL-bluff; blocked the green gate).

Root cause (FACT, not guess): `extension/tests/security/scanner-hostile-input.test.ts` asserted a tight absolute `expect(elapsed).toBeLessThan(5000)` on a 50k-anchor orchestrator scan. On a shared/oversubscribed host (load-avg 15) the *correct* scan measured **6608 ms**; in isolation it passes < 5000 ms (proven 5/5). Product behaviour was always correct (exactly 2 magnets detected, all junk excluded) — only the absolute timing tripped.

Affected files: `extension/tests/security/scanner-hostile-input.test.ts`.

Fix: replaced the tight 5000 ms budget with a generous **30 s hang-ceiling** (catches a true hang / quadratic explosion — a real $O(n^2)$ over 50k anchors takes minutes) and moved the rigorous machine-independent DoS guard to a new perf test. Correctness assertions retained.

Regression guard: new `extension/tests/perf/orchestrator-scaling.perf.test.ts` asserts a metamorphic sub-quadratic scaling ratio $(t(10 \cdot N) / t(N) < 30$; linear ~ 10 , quadratic ~ 100 — dimensionless, immune to host load) with a golden-good/golden-bad oracle self-validation ($\$11.4.107(10)$): `linearRatio=9.9` accepted, `quadraticRatio=113` rejected.

19. `crypto.perf.p99` budget — contention-coupled FAIL-bluff

Severity: MEDIUM (flaky test).

Root cause (FACT): `extension/tests/perf/crypto.perf.test.ts` asserted the 80 ms budget on `dist.p99.PBKDF2` — is CPU-bound; under concurrent-suite contention p99 spiked to **164 ms** while intrinsic cost stayed ~ 23 ms (proven in isolation). The newly-added heavy parallel perf/stress suites added the contention that exposed it.

Affected files: `extension/tests/perf/crypto.perf.test.ts`.

Fix: assert the budget on `dist.min` (intrinsic, contention-robust — contention only adds time). A real ~ 3.6 — algorithmic regression still inflates the min past 80 ms; correctness (EXACT plaintext recovery) is asserted every round.

Regression guard: the same test (now contention-robust); isolated min ~ 23 ms $\ll 80$; full-suite CI-EXT: PASS.

20. `parsers.perf` scaling-ratio — contention-coupled FAIL-bluff

Severity: MEDIUM (flaky test).

Root cause (FACT): `extension/tests/perf/parsers.perf.test.ts` computed the per-link scaling ratio from the **median** time at each size. The small (1000-link, ~ 5 ms) median is noise-sensitive under contention; the ratio tripped **3.09** > 3 while the intrinsic ratio is ~ 1.0 (proven 1.34 / 1.00 in isolation).

Affected files: `extension/tests/perf/parsers.perf.test.ts`.

Fix: compute per-link cost from the **min** run at each size (intrinsic). A genuine $O(n^2)$ regression still shows ratio ~ 5 in the min run, so the cap of 3 keeps full regression-catching power while never flaking.

Regression guard: the same test; isolated ratio 1.24 $\ll 3$; full-suite CI-EXT: PASS.

21. Three real WCAG 2.1 AA colour-contrast defects (popup theme tokens)

Severity: MEDIUM (accessibility â€” WCAG SC 1.4.3, real product defect).

Root cause (computed, Â§11.4.6): three popup theme tokens fell below the 4.5:1 normal-text contrast floor. Found by the new `extension/tests/a11y/focus-and-contrast.a11y.test.ts` (computes contrast ratios from the REAL committed CSS; analyzer self-validated against WCAG reference extremes), kept RED until fixed.

Affected files: `extension/src/popup/styles.css` (each token in both the `@media prefers-color-scheme` block and the explicit `[data-theme]` class).

Fix (same hue family, minimal visual change, â‰¥4.6:1): `--text-faint` dark on `#1a1a2e`: `#7a7a90` (4.07) â†’ `#838399` (4.61) `--text-faint` light on `#f5f6fa`: `#8a8a9c` (3.14) â†’ `#6c6c7e` (4.76) `--warning-text` light on `#fff6e0`: `#9a6a00` (4.40) â†’ `#946400` (4.78)

Regression guard: the 3 a11y tests flipped REDâ†’GREEN on the fix (proving they catch the defect) and are permanent Â§11.4.135 guards.

2026-06-13 â€” BobaLink extension wave-11: options-page WCAG a11y fixes

Found by the new `extension/tests/a11y/options-contrast-motion.a11y.test.ts` (computes contrast ratios from the REAL committed `src/options/styles.css`; analyzer self-validated against WCAG reference extremes), kept RED until fixed. Batch verified by `extension/ci-ext.sh` â†’ CI-EXT: PASS with the wave-11 coverage additions. All REDâ†’GREEN proven; the Â§11.4.120 reconciliation rewrote the gates to assert the NEW mechanism (not fake-passed, not reverted).

22. Save-button text below WCAG AA over its gradient (SC 1.4.3)

Severity: MEDIUM (accessibility, real). **Root cause (computed):** `.btn-primary` has `color:#fff` over `linear-gradient(--accent â†’ --accent-2)`; white on the lighter `--accent` (`#667eea`) end is only **3.66:1** (14px normal-weight text needs â‰¥4.5). **Fix:** the global `--accent` is ALSO `.nav-item.active` text on a dark sidebar where darkening it would REDUCE contrast, so the button uses a **button-local** darker indigo start `#5a6fce` (4.57:1); the `--accent-2` end is already 6.37:1. **Affected:** `src/options/styles.css` `.btn-primary`. **Guard:** the test now reads the buttonâ€™s ACTUAL gradient stops and asserts `#fff` â‰¥ 4.5 over every stop.

23 & 24. Field labels + nav-hover invisible in light theme â€” hard-coded literals (SC 1.4.3)

Severity: MEDIUM (accessibility, real â€” labels effectively unreadable in light mode). **Root cause (computed):** `.field > label` (`#d0d0e0`) and `.nav-item: hover` (`#e0e0e0`) were hard-coded **dark-theme** literals that never re-resolved per theme â†’ **1.40:1** and **1.12:1** on the light backgrounds. **Fix:** tokenized to theme-aware tokens â€” `label` â†’ `var(--text)` (10.6/13:1), `nav-hover` â†’ `var(--text-strong)` (15.5/16.1:1) â€” legible in both themes. **Affected:** `src/options/styles.css`. **Guard:** the Â§11.4.120-reconciled test asserts the

selectors now use a `var (--token)` (not a literal) AND the resolved token clears AA in BOTH themes.

25 & 26. No prefers-reduced-motion escape (SC 2.3.3 / 2.2.2)

Severity: MEDIUM (accessibility, real). **Root cause:** the options CSS ships `@keyframes fadeIn` (panel reveal on tab activation) + transitions, and the popup CSS ships transitions, but neither had a `@media (prefers-reduced-motion: reduce)` block — motion-sensitive users had no way to suppress it. **Fix:** added the standard reduced-motion block (zeroing animation/transition durations) to BOTH `src/options/styles.css` and `src/popup/styles.css`. **Guard:** the ally tests assert each stylesheet ships non-trivial motion AND a reduced-motion block (RED → GREEN on the fix).

2026-06-13 — BobaLink extension wave-12: scaling-ratio FAIL-bluff hardening

Batch verified by `extension/ci-ext.sh` — CI-EXT: PASS, full suite **799 passed (799)** (+38 over wave-11: crypto-tamper 17, link-scanner-coverage 10, highlight-manager 11 — all green, no product defects). The one fix below was a flaky test exposed by the heavier 799-test load.

27. infohash dedup scaling-ratio test — single-run + tiny-floor FAIL-bluff (Â§11.4.50)

Severity: MEDIUM (flaky test — Â§11.4.1 FAIL-bluff; blocked the green gate). **Root cause (FACT):** `tests/security/infohash-detection-hostile.test.ts`’s relative-scaling guard used a `SINGLE-run tLarge / Math.max(tSmall, 0.05)`; on a sub-ms baseline the 0.05 ms floor made the ratio noise-dominated (**53.6** observed under full-suite contention vs ~4 intrinsic). Worse, the ≈ 40 threshold was too loose to even catch the quadratic it guards (4 — input — linear⁴, quadratic¹⁶ — both <40). Passed at 761 tests, tripped at 799. **Fix:** measure the MIN over 7 reps at each size (intrinsic, contention-robust — host stalls only ADD time) and tighten the threshold to **10** (sits BETWEEN linear⁴ and quadratic¹⁶, so it now genuinely catches an $O(n^2)$ regression while never flaking). Verified stable 5/5 in isolation + full-suite CI-EXT: PASS. **Affected:** `extension/tests/security/infohash-detection-hostile.test.ts`. **Note:** the sibling stress test’s median estimator was initially assessed robust — but it ALSO flaked one wave later under heavier load (see entry 28), confirming that median is not enough; min is.

2026-06-13 — BobaLink extension wave-13: tab-group scaling median → min + working-tree hygiene

Batch verified by `extension/ci-ext.sh` — CI-EXT: PASS, full suite **814 passed (814)** (+15 over wave-12: popup-states 5 — genuine partial-Send-All gap; plus two prior-session test files brought into git — `options-save-flow4` + `offline-queue-persistence6` — that were sitting UNTRACKED in the working tree). No product defects.

28. tab-group scaling-ratio test “ median estimator spikes under load (11.4.50)

Severity: MEDIUM (flaky test “ 11.4.1 FAIL-bluff; blocked the green gate). **Root cause (FACT):** tests/stress/orchestrator-ratelimiter-tabgroup.stress.test.ts’s scaling guard used the MEDIAN of 5 reps; under the heaviest concurrent load (814 tests) several reps landed on a busy core and dragged the median up, so the 10—input ratio spiked past the 25 ceiling while the intrinsic ratio is ~5“8. (This refutes the wave-12 assessment that median was robust “ entry 27’s note.) **Fix:** switched the shared timing helper from median to **MIN** over reps (the minimum is the intrinsic-cost estimator “ host stalls only ADD time, so the fastest run is closest to true cost; a genuine $O(n^2)$ regression still inflates every run including the min) and bumped reps (orchestrator 3†’5, tab-group 5†’9). Verified stable 3/3 in isolation (ratios 4.76/6.41/8.25 vs the 25 ceiling) + full-suite CI-EXT: PASS. **Affected:** extension/tests/stress/orchestrator-ratelimiter-tabgroup.stress.test.ts. **Lesson (11.4.118):** the robust estimator for CPU-bound scaling-ratio tests under concurrent load is MIN-of-reps, not median/mean “ all six flaky-scaling fixes this session converge on it.

2026-06-13 “ BobaLink extension wave-14: live-7187 harness hardening

Live-harness prep to de-risk the operator-gated live round-trip. The default suite is unchanged at **814 passed (814)** (CI-EXT: PASS); the live test + challenge are out-of-suite and operator-run.

29. live_detect_send challenge “ 11.4.14 cleanup-race leaves an orphan synthetic torrent

Severity: MEDIUM (anti-bluff/hygiene defect in a live Challenge). **Root cause (FACT):** challenges/extension/live_detect_send_challenge.sh cleaned up the synthetic torrent it adds ONLY when its secondary qBittorrent torrents/info probe reported present. But the proxy adds the magnet itself; that separate probe can race/auth-fail and read absent while the magnet IS added “ so a PASS could leave an orphan synthetic torrent in qBittorrent (a 11.4.14 quiescence violation). **Fix:** cleanup now keys off the **proxy’s authoritative per-url verdict** (results[0].status == "added") OR the probe, re-logs-in for a fresh SID when needed, and warns honestly if delete-login fails (never reds the run at teardown). Also fixed a /bin/sh -n parse issue (apostrophe/metachar in added comments “ 11.4.67). **Affected:** challenges/extension/live_detect_send_challenge.sh. **Verification:** clean SKIP (exit 77) when the backend is down (no fail-open), bash -n + sh -n PASS, mutation-verified (no-op/partial response stubs †’ FAIL); the cleanup trigger confirmed to fire on a simulated proxy-added + probe-absent response (the old code would not). The up-path runs when the operator brings the stack up.

2026-06-13 “ Boba broader-project (merge-service / Go / plugins / frontend) verification + 2 fixes

A full-project verification pass (parallel subagents, all run against the REAL toolchains, anti-bluff captured evidence) confirmed the broader Boba backend is green: **Go qBittorrent-go 14 packages PASS under -race, vet clean, no data races; plugins: 60 .py compile, 12 curated**

contract-valid, 901 plugin tests pass (under `.venv/py3.13`); **Angular frontend: build GREEN + 342 unit tests pass**; **merge-service Python: 4149 unit tests pass, ruff clean** (under `.venv/py3.13`). Two genuine items found:

30. `ci.sh / scripts/run-tests.sh false-red` under a host `python3 < 3.12`

Severity: MEDIUM (the project's own manual CI gate false-reds a §11.4.1 FAIL-bluff at the toolchain layer). **Root cause (FACT, independently found by two subagents):** both scripts invoke a **bare** `python3`, which resolves to the host default. `pyproject.toml` declares `requires-python = ">=3.12"` and the suite uses `tomllib + PEP-604 X | Y` union syntax that a `python3 < 3.12` cannot even COLLECT. On this host `python3 = 3.9.25`, so `python3 -m pytest tests/unit/` aborts: **Interrupted: 36 errors during collection** (`TypeError: unsupported operand`), while the project `.venv (py3.13)` collects **4149** and passes all. **Fix:** both scripts now select a `>=3.12` interpreter via a `_select_python` helper (prefers `$PYTHON .venv/bin/python python3.13 python3.12` a bare `python3` only if it is `< 3.12`), aborting with a clear message if none qualifies. Backward-compatible: a host already on `python3 < 3.12` selects the same interpreter. **Affected:** `ci.sh` (lines for `py_compile` + the unit/integration/e2e pytest steps), `scripts/run-tests.sh` (the 5 `exec` | `pytest` modes). **Verification (captured):** `bash -n` clean both; the selector picks `.venv/py3.13`; bare `py3.9` **Interrupted: 36 errors during collection** vs selected interpreter **4149 tests collected** on the same `tests/unit/` dir.

31. Go backend **2 uncovered critical paths closed (coverage, not a defect)**

Type: Task (coverage). The `qBittorrent-go merge_search RunSearch` goroutine-orchestration loop (every prior test passed a `nil` client) and the `sse_broker` concurrent pub/sub had zero real coverage. Added 4 anti-bluff tests (`internal/service/coverage_test.go`, real `httptest` `qBittorrent` server, no mocks) `RunSearch` completes/accumulates + `StartFails` failed + `ContextCancel` abort, and an 8-publisher/16-subscriber `SSEBroker` churn under `-race`. **Anti-bluff proof:** a no-op `return nil` stub in `RunSearch` made all three `RunSearch` tests FAIL, then reverted (production clean, §11.4.84). Verified GREEN under `-race` (service package ok 5.933s, no data race). Production code unchanged.

2026-06-13 **Boba merge-service hermetic specialized suites: 5 test-quality fixes (product correct)**

A verification pass over the merge-service's hermetic specialized suites (`stress`, `chaos`, `property`, `contract`, `concurrency`, `memory`, `observability`, `security`) run under `.venv/py3.13`, the suites BG-1's `tests/unit` pass did not cover) found the PRODUCT is correct + secure (bandit 0 HIGH; CORS hardened; no product defect) but surfaced 5 real TEST-quality defects across the anti-bluff classes. All test-side only `download-proxy/src/` unchanged; each verified green under `.venv`.

32. Five merge-service test-quality defects (Â§11.4.1 / Â§11.4.3 / Â§11.4.50 / Â§11.4.120)

- `tests/stress/test_pipeline_stress.py` â€” Â§11.4.50 **flaky absolute-wall-clock threshold**. Gated on `max(latencies_ms) < 2000` (absolute); a lone GC/scheduler stall spiked one iteration to ~140 ms (~25 $\times$ the 5.6 ms median) â€” an absolute bound is the wrong oracle. Fixed to a RELATIVE bound `max(max(p50, 1.0 ms), max(latencies_ms, 200)) < 200` (an $O(n^2)$ blowup inflates the median too, so it still fires; a lone outlier no longer flakes). Verified: 3 passed.
- `tests/contract/test_crossapp_theme_contract.py` â€” Â§11.4.1/Â§11.4.3 **FAIL-instead-of-SKIP**. The 3 `@requires_compose` testsâ€™ own docstring promises â€œskip cleanly when the proxy is unreachableâ€”, but `_fetch()` did a bare `urlopen()` `URL error` `FAIL` when down. Fixed `_fetch()` to catch only `CONNECTION` errors `pytest.skip` (with `./start.sh -p reason`); `HTTPError` still returns the real status and `AssertionError` is NOT swallowed (a real missing-bridge still FAILs when the proxy is up). Verified: 3 SKIP when down, suite 27 passed when collectable.
- `tests/security/test_cors.py` â€” Â§11.4.120 **stale gate asserting REMOVED insecure behavior**. `TestCORSWildcardDefault` asserted `Access-Control-Allow-Origin == "*" as the default`, but the product was deliberately hardened to a secure-by-default localhost/dev allowlist (wildcard removed, CONTINUATION known-issue #5). The TEST lagged the security improvement. Reconciled to assert the NEW secure default â€” a known localhost origin reflected EXACTLY, an unknown LAN origin NOT reflected and never *. Anti-bluff: a wildcard-revert fails the `!= "*" assertions (RED-on-old/GREEN-on-new)`, not a tautology. Verified.
- `tests/observability/test_metrics_exist.py` â€” Â§11.4.3 **infra-FAIL instead of SKIP**. A `@requires_compose` test made an unconditional `httpx.get(:7187/metrics)` `ConnectError` `FAIL` when down. Added the existing `@merge_service_required` skip guard `SKIPs` cleanly. Verified.
- `tests/security/test_jackett_autoconfig_secrets.py` â€” **toolchain-path FAIL**. `subprocess.run(["bandit"])` raised `FileNotFoundError` (bandit is in `.venv/bin`, not on bare PATH). Added a `_resolve_bandit()` helper (prefers the running interpreterâ€™s bin dir, then `shutil.which`, SKIP if genuinely absent). Product clean: `.venv/bin/bandit` on `jackett_autoconfig.py` `0` HIGH findings. Verified.

2026-06-13 â€” Boba broken-doc-links + perf-test container-boot (the last test-type suites)

Ran the remaining test types (performance/benchmark/load/docs) under `.venv`. Benchmark: 9 infra-free pass (500 $\times$ â€”â€”4760 $\times$ threshold headroom â€” not flaky, correctly left unchanged) + 5 by-design error-not-skip infra-down (intentional). Load: only a Locust `locustfile.py` (no pytest tests). Two real items fixed (BUGFIXES 33, 34).

33. docs / test caught 8 broken internal markdown links + 1 test FAIL-bluff

Severity: LOW-MEDIUM (broken doc links + a Â§11.4.1 test FAIL-bluff). **Root cause + fix (FACT):** `tests/docs/test_no_broken_links.py` (resolves links relative to each docâ€™s own dir, the markdown convention) flagged 8 docs. Seven were genuine broken links â€” `docs/*_Summary.md` linked to siblings with a spurious `docs/` prefix (`[Architecture Summary]` (`docs/Architecture_Summary.md`) in a doc already

inside docs/ â†’ resolves to docs/docs/...). Fixed by dropping the prefix [] (docs/X) â†’] (X)); docs/Scripts_Summary.mdâ€™s [(scripts/pre_build_verification.sh) â†’] (../scripts/...) (the script is at repo-root). The 8th â€™ a research doc â€™ was a **TEST FALL-BLUFF**: the link extractor matched [\$ {tab.title}] (\${tab.url}) inside a **JS code block** (not a doc link). Fixed the extractor to STRIP fenced + inline code before matching. Anti-bluff verified (Â§1.1): a real broken link in **prose** still FAILs (the strip only removes code, not real links â€™ not a tautology). **Affected:** 7 docs/*_Summary.md (+regenerated .html/.pdf), tests/docs/test_no_broken_links.py. **Verified:** tests/docs â†’ 129 passed / 0 failed / 2 honest skip (mkdocs absent).

34. Performance test BOOTED CONTAINERS + hung 120 s on infra-down (Â§11.4.3 / no-boot)

Severity: MEDIUM (a test that actively boots infra + hangs). **Root cause (FACT):** tests/performance/test_concurrent_search.py::TestConcurrentSearch consumed a merge_service_live fixture that, on ports-down, ran podman/docker compose up -d then wait_for_port(timeout=120) â€™ pytest-timeout killed the wait â†’ 7 setup-ERRORs + 2 min wall-clock, and it was ACTIVELY booting containers. The suiteâ€™s siblings (stress/security/observability) all use the fast merge_service_required probe-and-SKIP guard. **Fix:** applied that same merge_service_required guard (1 s socket+/health probe, never boots containers) â†’ 7 honest infra-down SKIPs. Non-tautology: when the stack is up the class still runs every real assertion (status 200, latency ceilings, p50/p95) unchanged. (The benchmark suiteâ€™s separate error-not-skip design is INTENTIONAL and was left untouched per Â§11.4.122/Â§11.4.6.) **Affected:** tests/performance/test_concurrent_search.py. **Verified:** 7 SKIP cleanly, no container boot, download-proxy/src/ unchanged.

2026-06-13 â€™ start.sh boot completely broken: boba-ctl up -d flag mismatch

35. ./start.sh aborted EVERY boot at the container-start step (-d flag)

Severity: HIGH (the documented startup path ./start.sh did not work at all). **Root cause (FACT, found by actually running the boot):** start.sh::start_container() runs the generic \$COMPOSE_CMD up -d. In the default boba-ctl mode \$COMPOSE_CMD = scripts/boba-ctl.sh, whose up|down) case did exec "\$BOBA_CTL_BIN" "\$@" â€™ passing the compose-style -d (detach) straight to the cmd/boba-ctl/boba-ctl Go binary, whose up defines only -profile/-wait. Result every boot: flag provided but not defined: -d â†’ [ERROR] Failed to start container â†’ exit 1, before any container was created. **Fix:** scripts/boba-ctl.shâ€™s up|down) case now strips -d/--detach (boba-ctl runs detached by default) so it is a true drop-in for compose up -d. **Affected:** scripts/boba-ctl.sh. **Verified:** bash -n clean; the re-run boot proceeded past the step into podman-compose ... up -d (created pod_boba, pulled all images).

Operator-environment note (NOT a project defect, reported per Â§11.4.6): with the -d bug fixed, the boot then HUNG inside podman-compose itself â€™ it created pod_boba + pulled every image, then froze at 0.0% CPU (STAT=SN) before creating any container. podman itself works (podman run alpine â†’ OK), so the hang is specific to podman-compose, which on this host runs under homebrew python@3.14 (a likely-incompatible combo; podman-compose support for 3.14 is not established). Native podman compose just delegates to the

same podman-compose. This is an operator-environment toolchain issue (fix: run podman-compose under a supported python, e.g. 3.11/3.12, or repair the podman-compose install), surfaced to the operator as the project's start.sh orchestrator code is now correct.

Clarifying footnote (2026-06-13, root-caused as superseded the python@3.14 hypothesis above): the hang was NOT python@3.14. Captured live: the vfkit VM shares are /Users, /private, /var/folders and podman machine inspect shows Mounts:[] as the repo lives on /Volumes/T7 (external SSD), which is NOT shared into the podman applehv VM, so the first podman create stalls on an unreachable bind path. Proven by both python 3.13 AND 3.14 hanging identically (refuting the version theory). Fix (operator-authorized, 11.4.122): recreated the machine with --volume /Volumes/T7:/Volumes/T7 --cpus 4 --memory 6144; all 5 containers then booted healthy. On macOS the network_mode: host ports live in the VM, so an SSH -L tunnel (or scripts/ensure-macos-tunnel.sh) bridges 7186/7187 to the Mac.

2026-06-13 as qBittorrent 5.x API-compat: download round-trip broken (login + add)

Surfaced by the live detect-send-torrent round-trip (extension/tests/live/download-endpoint.live.test.ts) once the stack was finally booted (qBittorrent v5.2.1, linuxserver/qbittorrent:latest). Three real PRODUCT defects + two test/challenge-quality items, each proven with captured live evidence. The whole class: the proxy + tests spoke legacy qBittorrent's plain-text protocol (200 "Ok.") while the image speaks 5.x's JSON/204.

36. Proxy reported auth_failed for EVERY download as modern qBittorrent 204 login

Severity: CRITICAL (no download could reach qBittorrent). **Type:** Bug. **Root cause (FACT, captured live):** download-proxy/src/api/routes.py judged WebUI login with resp.status == 200 and body == "Ok.". Modern qBittorrent (4.6+/5.x) returns 204 No Content with an EMPTY body + the QBT_SID cookie (HTTP/1.1 204 / set-cookie: QBT_SID_7185= as!) as status != 200 AND body != "Ok." as {"status": "auth_failed"} even though login succeeded. The defect lived at FOUR login call-sites (/download, /download/upload, /downloads/active, /auth/qbittorrent) as the active-downloads dashboard widget + credential-save endpoint were broken too. **Fix:** new_qbit_login_succeeded(status, body, cookies) as authoritative version-independent signal is as the server issued a QBT_SID/QBT_SID_<port> session cookie (exact/prefix match, not a loose "SID" substring), legacy Ok. body as fallback; all four call-sites migrated. **Affected:** download-proxy/src/api/routes.py. **Regression guard (11.4.135):** tests/unit/test_qbit_login_compat.py incl. test_reproduces_pre_fix_defect_old_check_rejected_modern_204 (11.4.115 polarity). **Verified live:** POST /api/v1/auth/qbittorrent admin/admin as {"status": "authenticated", "version": "v5.2.1"}; the round-trip went from auth_failed-SKIP to GREEN.

37. Proxy reported failed for successful adds as modern qBittorrent JSON add + duplicate 409

Severity: CRITICAL (a torrent that actually landed was reported failed). **Type:** Bug. **Root cause (FACT, captured live):** the add-success check was body.lower().startswith("ok").

Modern qBittorrent /api/v2/torrents/add returns a JSON summary, not Ok.:
 {"added_torrent_ids":
 ["<hash>"], "failure_count":0, "pending_count":0, "success_count":1}
 " the torrent IS added (confirmed in /torrents/info) but the proxy saw { ' failed
 with the JSON as detail. SECOND, related: BobaClient retries on network/timeout; attempt 1
 added the magnet but was slow ' client timed out ' retry re-sent the SAME infohash '
 qBittorrent 409 Conflict (duplicate) ' failed (proven: one round-trip = TWO /api/
 v1/download requests in the proxy log). **Fix:** new_qbit_add_succeeded(status,
 body) " accepts the modern JSON (added_torrent_ids non-empty OR
 success_count/pending_count >= 1, with defensive int-coercion so a malformed "N/A"
 count classifies as failure, never crashes the add path), legacy Ok., AND treats 409 Conflict
 as idempotent SUCCESS (the torrent IS present " makes a retried add safe); all three add call-
 sites migrated. **Affected:** download-proxy/src/api/routes.py. **Regression guard:**
 tests/unit/test_qbit_login_compat.py (modern-JSON / pending-only / all-failed /
 409-duplicate / non-numeric-count / legacy-Ok / Fails / 415 cases). **Verified live (3,**
11.4.50): the live round-trip + challenges/extension/
 live_detect_send_challenge.sh both GREEN " the synthetic infohash is
 INDEPENDENTLY confirmed present in qBittorrent's real /torrents/info, then
 cleaned up (11.4.14).

38. Live test + challenge could not cross-check modern qBittorrent (test-quality, no product defect)

Severity: MEDIUM (anti-bluff held: tests SKIP™d honestly, never false-PASS™d). **Type:** Task. **Root cause:** (a) the live test™s own confirm-login helper had the same legacy status==200 && "Ok." assumption ' could not read /torrents/info on 204 ' honest SKIP. (b) the challenge cross-checked qBittorrent at :7185 (container-internal, not reachable off-host on macOS). **Fix:** (a) confirm-login now accepts 200/204 + keys on the Set-Cookie SID (mirrors the proxy); (b) challenge default qBittorrent base :7185 ' :7186 (the documented WebUI access port, reachable on Linux + macOS). **Affected:** extension/tests/live/download-endpoint.live.test.ts, challenges/extension/live_detect_send_challenge.sh. **Verified:** both GREEN with real infohash-present confirmation + 11.4.14 cleanup.

Code review (11.4.142/11.4.134): an independent adversarial review caught 2 BLOCKING (2 of the 4 login sites initially missed; an int("N/A") crash path) + 1 warning (loose "SID" substring) " all remediated + guarded; the re-review returned a clean GO.

2026-06-13 " merge dashboard: search-result buttons dead during search + results collapse on completion

Operator-reported (searched "Linux", clicked buttons): "Magnet/Download/qBit buttons don't work while the search is running; after completion they work but the results are just a fraction of what we saw streaming." Root-caused via systematic-debugging (11.4.102), fixed TDD (RED proven, 11.4.115), independent code review GO (11.4.142/11.4.134), live-verified on the running stack.

39. Search-result buttons (Magnet / Download / qBit) were a NO-OP while a search was in progress

Severity: HIGH (core user action unusable mid-search). **Type:** Bug. **Root cause (FACT, code evidence):** the Angular dashboard grid renders *ngFor="let r of sortedResults()";

let `i = index` and the buttons call `doMagnet(i)/doSchedule(i)/doDownload(i)`, but the handlers read `this.results()[i]`. `results()` is EMPTY during a search (only `loadSearchResults()` sets it on `search_complete`) and in a DIFFERENT order after `â†’` while the grid shows `sortedResults()` (`= liveResults()` during search). So mid-search `results()[i]` is undefined `â†’ if (!r) return â†’` every button did nothing; after search the index desynced from the sorted display (wrong-row risk). **Fix:** handlers resolve the row from `this.sortedResults()[index]` `â†’` the SAME array the template iterates. Plus the deferred `qBit-login` callback now captures the ROW object (`executeScheduleRow(r)`) instead of re-resolving by index, eliminating a search-completes- mid-dialog wrong-row race (review nit). **Affected:** `frontend/src/app/components/dashboard/dashboard.component.ts`. **Regression guard (Â§11.4.115/Â§11.4.135):** `dashboard.component.spec.ts` `â†’` 4 new tests (each button fires for the live row mid-search + the deferred-login row-capture); RED-proven (pre-fix `â†’` `â†’` expected one matching request, found none`â†’`). **Verified live:** `challenges/extension/search_buttons_live_challenge.sh` PASS `â†’` all 3 button backends (`/api/v1/magnet`, `/api/v1/download/file`, `/api/v1/download`) satisfy their real contract; the synthetic magnet is added + INDEPENDENTLY confirmed present in `qBittorrent` + cleaned up (Â§11.4.14). Full frontend suite 345 green.

40. Search results collapsed to a fraction (50) on completion despite thousands streamed

Severity: HIGH (perceived data loss). **Type:** Bug. **Root cause (FACT, code + runtime evidence):** `get_search` (the endpoint the grid swaps to on `search_complete`) hardcoded for `m in merged[:50]`: (not even the requested `limit`). Captured live: a `â†’Linuxâ†’` search streamed **2153 results** into the live grid, but the final list was capped at **50** `â†’` the operator`â†’` `â†’` just a fraction.`â†’` **Fix:** `get_search(search_id, req, limit: int = 0)` returns the FULL merged set by default (`display = merged if limit <= 0 else merged[: max(0, limit)]`); the grid uses `cdk` virtual-scroll so the full set renders cheaply. **Affected:** `download-proxy/src/api/routes.py`. **Regression guard (Â§11.4.115/Â§11.4.135):** `tests/unit/test_search_results_not_capped.py` (3 tests: 73 merged `â†’` all 73 not 50; optional `?limit` truncates; sub-50 returned whole) `â†’` RED-proven (cap-50 `â†’` 50% 73). **Verified live (Â§11.4.123):** post-fix `â†’Linuxâ†’` search `â†’` completed `total=1363 merged=862 RESULTS_LEN=862` (was 50). The final set now matches the merged count.

41. pytest tore down the operator`â†’` running stack on every suite that touched `docker_services`

Severity: MEDIUM (test-infra; repeatedly killed the live stack mid-session). **Type:** Bug. **Root cause (FACT):** `tests/conftest.py` overrode `docker_setup` `â†’` `[]` (skip `compose-up`) but NOT `docker_cleanup`, so `pytest-docker` `â†’` default teardown ran `compose down`, destroying a stack the suite did NOT bring up. Any suite using the `docker_services` fixture (e.g. `â†’` the contract/security live-probing tests) nuked `qbittorrent` + the proxy on teardown. **Fix:** add a `docker_cleanup` fixture returning `[]` (mirrors `docker_setup`) `â†’` the suite is now a pure observer; startup/shutdown is owned by `./start.sh`. **Affected:** `tests/conftest.py`. **Composes Â§11.4.119** (single-resource-owner `â†’` tests must not seize a resource in use).